

AVA-v2: Reasoning on a 4 GB Laptop GPU

Muhammad Afsah Mumtaz

Independent Researcher

github.com/NAME0x0 huggingface.co/NAME0x0

Abstract

Open language models have outgrown the hardware most people own. We report an end-to-end case study of adapting and evaluating a compact model within a strict consumer budget. Beginning from Qwen3.5-2B, an approximately two-billion-parameter vision-language model, we fine-tune only its text transformer with 4-bit QLoRA on a single laptop GPU that has 4 GB of memory. Training peaks at 1.81 GB of VRAM, runs for 100 minutes, updates 0.58% of the parameters, and produces a 42 MB adapter. We evaluate the merged model, which we call AVA-v2, across 17 benchmark configurations covering knowledge, reasoning, mathematics, code, multilingual arithmetic, and instruction following, and report a 95% Wilson interval for every score. The capability profile is uneven: strong on science and multiple-choice knowledge (ARC-Challenge 82.0%, ARC-Easy 92.0%, MMLU 59.2%) and weak on competition mathematics and strict instruction following. Two results are worth isolating. A 50-question GSM8K estimate of 48% overstated the full 1,319-question result of 35.3% by about thirteen points, which shows how small subsets can misrepresent compact models. Self-consistency added 8.7 points on GSM8K, whereas an in-model tool-use pathway activated on 0.6% of problems and changed nothing. The adapter, merged weights, and a quantized GGUF ladder are publicly available.

Index Terms—QLoRA, parameter-efficient fine-tuning, small language models, low-resource training, model evaluation, quantization.

I. INTRODUCTION

The capability of open language models has grown faster than the hardware most people can run them on. Current systems assume several high-memory accelerators, yet many students, hobbyists, and independent researchers have access to a single consumer GPU with 4 to 8 GB of memory. What is achievable at the low end of that range is rarely documented in full. This paper works through one instance from start to finish: the adaptation, evaluation, and deployment of a compact model on a laptop GPU with 4 GB of VRAM.

We begin from Qwen3.5-2B, a recent vision-language model of roughly two billion parameters [20], and adapt only its text transformer with 4-bit QLoRA [8, 13]. The run peaks at 1.81 GB of VRAM, completes in 100 minutes, and writes a 42 MB adapter that modifies 0.58% of the parameters. We refer to the merged result as AVA-v2. Training and inference both stay under 2 GB, so the model runs on the same hardware that produced it.

Rather than report a single favorable number, we evaluate AVA-v2 across 17 benchmark configurations and attach a 95% Wilson interval to each score. The profile that emerges is mixed, and in places unflattering. That is the point. A small model adapted cheaply is good at some things and poor at

others, and describing that precisely is more useful than a headline.

One observation during the evaluation has consequences past this particular model. An early estimate on 50 GSM8K questions reported 48% accuracy. The full 1,319-question test set reports 35.3%. The convenient subset overstated the real figure by about thirteen points, a gap several times larger than the confidence interval on either measurement.

We organize the study around three questions.

- 1) *Feasibility*. Can a two-billion-parameter model be adapted to a useful degree under 2 GB of peak VRAM on consumer hardware?
- 2) *Capability profile*. Which abilities survive or improve after low-resource QLoRA adaptation, and which stay weak?
- 3) *Evaluation reliability*. How far can small benchmark subsets mislead when estimating the performance of a compact fine-tuned model?

The paper makes four contributions.

- 1) An adaptation pipeline that fits a two-billion-parameter model into 1.81 GB of peak training VRAM on a 4 GB laptop GPU, with every resource cost reported.
- 2) A broad evaluation across 17 benchmark configurations with Wilson intervals, including execution-based coding and self-consistency.
- 3) Evidence, with a worked example, that small evaluation subsets can substantially misstate the performance of compact models.
- 4) Public release of the adapter, merged weights, and an imatrix-calibrated GGUF quantization ladder with measured perplexity.

A caveat frames everything that follows. The base model is a capable recent system, and the adaptation described here does not make it broadly stronger; on several general benchmarks a well-run base would likely score higher. The value of this study is the reproducible low-resource process, the full-set measurements, and what those measurements reveal about evaluating small models, not a claim that AVA-v2 surpasses its base or any larger model.

II. RELATED WORK

a) *Parameter-efficient and quantized adaptation.*: LoRA freezes the pretrained weights and learns low-rank updates to selected projections [13]. QLoRA quantizes the frozen base to 4-bit NF4 and back-propagates through it into the LoRA parameters, cutting the memory needed to fine-tune a billion-parameter model by roughly an order of magnitude [8].

Recent surveys place these methods within a wider family of parameter-efficient techniques for large models [19]. We use this recipe without modification. The aim is not a new adaptation method but a complete account of what the standard one costs and yields at the 4 GB limit.

b) Compact and on-device models.: A separate line of work targets models small enough to run locally, either by pretraining at reduced scale or by curating higher-quality data [30, 10]; recent surveys catalog these model families and their measured accuracy and efficiency trade-offs [17]. AVA-v2 is not a new base model. It is an adaptation of an existing 2B model, studied under the constraint that the entire process, training through evaluation, happens on one laptop GPU.

c) Quantized local deployment.: The llama.cpp project and the GGUF format have made low-VRAM and CPU inference of quantized models routine [9]. Quality at aggressive quantization is often assumed rather than measured. We give perplexity across an importance-matrix-calibrated ladder so the deployment trade-off is explicit.

d) Reliability of benchmark evaluation.: Scores for small models are sensitive to prompt format, few-shot selection, and sample size [2], and to overlap between training and evaluation data [23]. We add a concrete case in which a 50-item subset overstated full-set GSM8K accuracy by thirteen points, reported with confidence intervals sized to make such gaps visible.

e) Inference-time reasoning.: Chain-of-thought prompting [26] and self-consistency [24] raise reasoning accuracy without retraining. We measure the self-consistency gain for AVA-v2 and compare it against an in-model tool-use pathway that rarely activates, and find the former far more effective for this model.

III. LOW-RESOURCE TRAINING SYSTEM

A. Base model

The base model is Qwen3.5-2B, a vision-language model released by the Qwen team under an Apache-2.0 license [20]. It pairs a text transformer with a vision encoder through early-fusion multimodal training, and its language stack combines Gated DeltaNet layers, gated attention, and a sparse mixture-of-experts feed-forward network, with a native context of 262,144 tokens. The checkpoint contains 1,892,736,832 parameters in total.

We adapt only the text transformer. The vision encoder is left untouched and is never exercised: every dataset in this study is text-only, and the model is prompted purely as a language model. The description that follows therefore concerns the language stack alone.

B. Adaptation

We fine-tune with QLoRA [8], which keeps the base weights frozen and quantized to 4-bit NF4 while training low-rank adapters on top [13]. Adapters of rank 16 with scaling factor $\alpha = 32$ and no dropout are attached to the query, key, value, output, gate, up, and down projections of every transformer block. This leaves 10,911,744 parameters trainable, 0.58% of

TABLE I
TRAINING CONFIGURATION. THE CORPUS AND ITS PROVENANCE ARE DESCRIBED IN SECTION IV.

Setting	Value
Base model	Qwen3.5-2B (text transformer)
Total parameters	1,892,736,832
Trainable parameters	10,911,744 (0.58%)
Base quantization	4-bit NF4
LoRA rank / α / dropout	16 / 32 / 0
Target modules	q, k, v, o, gate, up, down projections
Attention	scaled dot-product
Sequence length	384
Batch size $\times$ grad. accumulation	1×8
Epochs / optimizer steps	1 / 2,593
Learning rate	1.5×10^{-4}
Training examples	20,741

the model. Attention uses a scaled dot-product implementation. Table I lists the configuration.

C. Fitting the run into 4 GB

Several choices keep the run inside the memory of a single RTX A2000 Laptop GPU with 4 GB of VRAM, on Windows 11. The base is quantized to 4-bit, so the frozen weights occupy roughly 1 GB rather than the four they would take in half precision. A short sequence length of 384 tokens caps the activation and attention-cache footprint, which is the term that grows with context. A batch size of one with gradient accumulation over eight steps gives an effective batch of eight without holding eight sequences in memory at once. Periodic in-loop evaluation is disabled: the model has a large vocabulary, and materializing logits over it for an evaluation batch overflows the remaining memory, so we hold out 200 examples but score them after training rather than during it.

Under these settings the run peaks at 1.81 GB of VRAM, which leaves headroom on a 4 GB card and keeps inference on the same hardware comfortable. Training a single epoch of 2,593 steps takes 100.5 minutes and ends at a training loss of 0.4145. The resulting adapter is 41.6 MiB on disk.

IV. DATA AND CONTAMINATION

The corpus is a mixture assembled to strengthen behavior on public multiple-choice and grade-school mathematics tasks while preserving the tool-use and identity behavior of the assistant. Table II lists its composition. The science set draws multiple-choice questions from the SciQ [27] and OpenBookQA [18] training splits, rewritten as short explained answers. The mathematics set pairs GSM8K training questions [7] with chain-of-thought solutions. The ARC-Challenge set takes the training split [6] and augments each question with rotated answer-choice orderings, which is why 1,119 training questions expand to 4,476 examples. The remainder is a small amount of distilled mathematical reasoning and a few dozen hand-written examples covering calculator and code tool use, short reasoning puzzles, assistant identity, and everyday dialogue.

TABLE II

TRAINING CORPUS COMPOSITION. EVERY LEARNED COMPONENT COMES FROM A TRAINING SPLIT OR IS WRITTEN BY HAND; NO VALIDATION OR TEST SPLIT OF ANY EVALUATED BENCHMARK CONTRIBUTES TO IT.

Component	Examples	Source	Split
Science multiple-choice	8,000	SciQ, OpenBookQA	train
GSM8K chain-of-thought	7,473	GSM8K	train
ARC-Challenge, choice-augmented	4,476	ARC-Challenge	train
Distilled math reasoning	900	teacher distillation	—
Hand-written examples	37	synthetic	—
Tool-use augmentation	55	synthetic	—
Total before holdout	20,941		
Held out, unused	200		
Used for training	20,741		

A. Contamination check

Because the model is adapted on the ARC-Challenge and GSM8K training splits and later evaluated on their test splits, we measured overlap directly. For each evaluation set we normalized question stems, lowercasing and stripping punctuation and the answer-option block, and compared them against the training corpus by exact match and by 13-gram overlap. GSM8K showed no exact overlap across its 1,319 test questions. ARC-Challenge showed 8 exact stem matches out of 1,172 (0.68%). The remaining fifteen benchmarks share no data with the corpus by construction, since their datasets are absent from it.

The ARC-Challenge matches are a property of that benchmark rather than an artifact of the data pipeline. ARC’s own training and test splits repeat a small number of templated science questions, for example "which of these is a function of all cells," so a handful of stems in the test set also appear in the training split we adapted on. Even in the worst case, that all eight were answered from memorization, removing them lowers ARC-Challenge accuracy from 82.0% to 81.9%, inside the confidence interval reported in Section VI. Surveys of contamination and membership inference describe the general form of this risk [23].

One point should be explicit. Adapting on the ARC-Challenge and GSM8K training splits is a deliberate choice, and it means the model has seen in-distribution examples of these task formats. The scores reported here are held-out test results, not evidence of zero-shot generalization to unfamiliar task types.

V. EVALUATION METHODOLOGY

A. Benchmarks

We evaluate across six capability families. Knowledge and commonsense multiple choice covers ARC-Easy and ARC-Challenge [6], MMLU [11] and MMLU-Pro [25], HellaSwag [29], PIQA [3], WinoGrande [21], BoolQ [5], and TruthfulQA [15]. Mathematical reasoning covers GSM8K [7], MATH-500, a 500-problem subset of the MATH dataset [12] introduced by Lightman et al. [14], and the English, Spanish, and French portions of MGSM [22]. Code generation uses HumanEval+ and MBPP+, the test-hardened variants of HumanEval [4]

and MBPP [1] from EvalPlus [16]. Instruction following uses IFEval [31]. We also run GSM8K under self-consistency and under an agentic tool-use harness. MMLU uses five in-context demonstrations; the other multiple-choice sets are zero-shot.

B. Serving

All scores use the merged model exported to a Q8_0 GGUF and served with llama.cpp [9] (build 69c28f1) through its server, with flash attention, an 8-bit key-value cache, four parallel slots, continuous batching, and an 8,192-token context. The 8-bit export retains essentially all of the model’s quality; Section VIII reports the perplexity cost of heavier quantization.

C. Scoring

Multiple-choice questions are scored by a single forward pass: we read the next-token probabilities restricted to the candidate label tokens and take the arg-max. This differs from the log-likelihood-of-continuation scoring used by some harnesses, in which each full answer string is scored and the most likely is chosen. Letter arg-max measures the answer the model would actually emit; its absolute values are directionally comparable to leaderboard numbers but not identical to them.

Generative tasks use greedy decoding with the model’s chat template and its non-thinking mode. Self-consistency draws five samples at temperature 0.7 with nucleus sampling and takes the majority vote over the extracted numeric answer [24]. Code tasks run each generated program in a subprocess sandbox with a five-second limit and a 4 GB memory cap, and count a problem solved only when it passes the full EvalPlus test suite. IFEval checks each response against its instruction constraints with a rule checker covering roughly twenty-five instruction types, and reports prompt-level strict accuracy. The agentic harness exposes a calculator and a Python interpreter and allows up to three tool rounds per question.

D. Confidence intervals

Every accuracy is reported with a 95% Wilson score interval [28]. For k successes in n trials with $\hat{p} = k/n$ and $z = 1.96$, the interval is

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}.$$

The Wilson interval stays well behaved at the small sample sizes and extreme proportions that a compact model produces, where the normal approximation does not.

E. Accounting

We report 17 benchmark configurations comprising 52,027 evaluation instances. The number of model calls is slightly higher, about 52,045, because self-consistency issues five generations per question and a small number of items were retried. The count of distinct underlying questions is lower still, since GSM8K appears in four configurations.

VI. RESULTS

Table III reports all 17 configurations with 95% Wilson intervals.

The model is strongest on knowledge and common-sense multiple choice. ARC-Easy reaches 92.0% and ARC-Challenge 82.0%, and MMLU reaches 59.2% at five shots. On MMLU, 373 of 14,042 questions (2.7%) exceeded the 8,192-token context once five demonstrations were prepended, and were scored as incorrect; on the questions that fit, accuracy is 60.8%. BoolQ (75.0%) and PIQA (75.9%) are close behind, while HellaSwag (56.8%), WinoGrande (56.4%), and TruthfulQA (47.5%) are weaker. MMLU-Pro, which is harder and zero-shot, falls to 30.9%.

Mathematics is the weakest area. GSM8K is 35.3% under greedy decoding and 44.0% under five-sample self-consistency. MATH-500 is 18.8%, and MGSM averages 34.4% across three languages. Code generation is functional but modest: 19.5% on HumanEval+ and 35.7% on MBPP+. IFEval strict accuracy is 31.6%. The agentic configuration scores 35.4% on GSM8K, indistinguishable from the greedy run; Section VII examines why.

A controlled comparison against the unadapted base model is not available. An earlier partial evaluation of the base, on small samples and under a different quantization, placed it near 66% on ARC-Challenge and 28% on GSM8K, but those figures use a different protocol and sample size, and a full-set re-evaluation of the base is still pending. We therefore treat the base only as loose context and make no claim of superiority over it. Published scores for other 2B-class models are likewise measured under different prompting, shot counts, and scoring, and are not directly comparable to the values reported here.

VII. CAPABILITY AND FAILURE ANALYSIS

A. Where knowledge is strong and weak

Table IV breaks MMLU down by subject. The strongest areas are marketing (84.6%), management (83.5%), and high-school biology (81.0%). The weakest are advanced mathematics and computer science, alongside moral scenarios, a subject known to depress scores across models. MMLU-Pro shows the same shape by category: biology reaches 64.3%, while mathematics (21.4%) and law (20.3%) sit near what ten-option chance and format sensitivity allow. The model carries broad factual and social-science knowledge and loses ground exactly where problems require multi-step symbolic work.

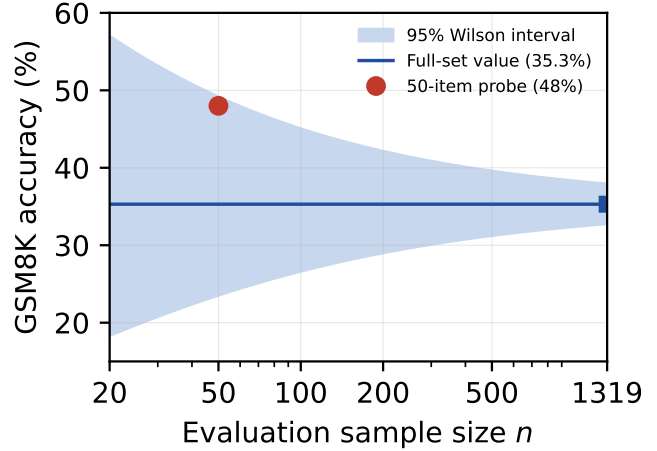


Fig. 1. The 95% Wilson interval for GSM8K accuracy around the full-set value of 35.3%, as a function of sample size. At $n = 50$ the interval spans roughly 22 to 47%, so the 50-item probe reading of 48% is a high-variance draw that the full test set does not support.

B. Mathematics degrades with difficulty and language

Two gradients make the mathematical weakness concrete. On MATH-500 (Table V), accuracy falls from 51.2% on the easiest problems to 5.2% on the hardest. On MGSM (Table VI), the same grade-school problems drop from 42.8% in English to 32.0% in Spanish and 28.4% in French, so the arithmetic reasoning does not transfer cleanly across languages.

C. Tool use rarely fires; self-consistency pays

The agentic harness changed nothing on GSM8K. The model emitted a tool call on only 8 of 1,319 problems (0.6%), of which 5 were correct, leaving accuracy at 35.4% against the 35.3% greedy baseline. The fine-tuning corpus held on the order of fifty tool-use examples against more than fifteen thousand worked math problems, so the model defaults to solving arithmetic directly rather than invoking the calculator. Self-consistency, by contrast, added 8.7 points on GSM8K for the cost of five samples and no retraining. For this model, sampling several chains of thought is the cheaper reasoning lever.

D. Small subsets misstate performance

The clearest methodological finding concerns evaluation size (Figure 1). An early estimate of GSM8K accuracy on 50 questions read 48%. The full 1,319-question test set reads 35.3%, with a 95% interval of [32.8, 38.0]. The subset overstated the full result by about thirteen points, well outside the interval of either measurement, because at $n = 50$ the interval on a proportion near one half spans roughly ± 14 points. For a compact model whose accuracy sits in the middle of the range, where binomial variance is largest, a fifty-item probe carries little information about the full-set number. This is a specific case of a general caution: reported gains for small

TABLE III
AVA-v2 ACROSS 17 BENCHMARK CONFIGURATIONS. MULTIPLE-CHOICE TASKS USE LETTER ARG-MAX; GENERATIVE TASKS USE GREEDY DECODING, EXCEPT THE SELF-CONSISTENCY ROW. ALL RUNS USE THE Q8_0 GGUF.

Benchmark	n	Acc. (%)	95% CI
ARC-Challenge	1,172	82.0	[79.7, 84.1]
ARC-Easy	2,376	92.0	[90.8, 93.0]
MMLU (5-shot)	14,042	59.2	[58.4, 60.1]
MMLU-Pro	12,032	30.9	[30.1, 31.8]
HellaSwag	10,042	56.8	[55.8, 57.8]
PIQA	1,838	75.9	[73.9, 77.8]
WinoGrande	1,267	56.4	[53.7, 59.1]
BoolQ	3,270	75.0	[73.5, 76.5]
TruthfulQA-MC1	817	47.5	[44.1, 50.9]
GSM8K	1,319	35.3	[32.8, 38.0]
GSM8K self-cons. ($k=5$)	200	44.0	[37.3, 50.9]
MATH-500	500	18.8	[15.6, 22.5]
MGSM (en/es/fr)	750	34.4	[31.1, 37.9]
HumanEval+	164	19.5	[14.2, 26.2]
MBPP+	378	35.7	[31.1, 40.7]
IFEval	541	31.6	[27.8, 35.6]
Agentic GSM8K	1,319	35.4	[32.9, 38.0]

TABLE IV
MMLU ACCURACY BY SUBJECT, STRONGEST AND WEAKEST TEN (5-SHOT).

MMLU subject	n	Acc. (%)
<i>Strongest 10</i>		
marketing	234	84.6
management	103	83.5
high school biology	310	81.0
high school psychology	545	80.5
international law	121	80.2
sociology	201	79.6
high school world history	237	78.1
high school government and politics	193	76.7
college biology	144	75.0
us foreign policy	100	75.0
<i>Weakest 10</i>		
virology	166	47.6
machine learning	112	45.5
college computer science	100	45.0
high school mathematics	270	43.3
college physics	102	43.1
professional law	1530	43.0
abstract algebra	100	43.0
moral scenarios	895	38.8
college mathematics	100	36.0
global facts	100	35.0

models should be read against the sample sizes that produced them.

VIII. DEPLOYMENT

AVA-v2 is released in three forms: the 41.6 MiB QLoRA adapter, a merged half-precision checkpoint for use with standard transformers, and a ladder of GGUF quantizations for CPU and low-VRAM inference through llama.cpp [9], Ollama, and LM Studio. Table VII gives the size and measured perplexity of each quantization.

The sub-8-bit files are built with an importance matrix calibrated on the model’s own training distribution, which keeps their quality close to the 8-bit reference. The recommended

Q4_K_M build is 1.19 GB and raises perplexity by 0.25% over Q8_0, while the smallest build fits in about 1.6 GB of memory. Every quantization runs without a GPU, so the model deploys on the same class of hardware that trained it. The evaluation in Section VI uses the Q8_0 build, the most faithful of the ladder.

IX. LIMITATIONS

Several limitations bound these results. The most important is the absence of a controlled comparison against the base model: the base was measured only on small samples under a different quantization, and a matched full-set re-evaluation is still pending, so we cannot say how much of the capability profile is due to adaptation rather than to the base itself. The study is also a single run. One seed, one corpus, and one 4 GB laptop GPU produced every number reported here, and we do not characterize variance across seeds, corpora, or hardware.

Two design choices limit scope. Training used a 384-token sequence length, which caps what the model can learn about long-context tasks, and the corpus held only a few dozen tool-use examples, which is why the agentic pathway almost never fires. The evaluation has its own edges. Multiple-choice scoring uses letter arg-max rather than log-likelihood of continuation, so absolute values are not directly comparable to leaderboards that use the latter, and 2.7% of MMLU questions overflowed the 8,192-token context at five shots and were counted wrong, which understates the five-shot number relative to a longer-context run.

Finally, the model was adapted on the ARC-Challenge and GSM8K training splits, so its scores on those benchmarks reflect in-distribution adaptation rather than zero-shot generalization, and the base is a capable recent system that the adaptation specializes rather than broadly improves.

X. CONCLUSION

A 1.9B model’s language backbone can be adapted, evaluated across seventeen benchmarks, and packaged for local

TABLE V
MATH-500 BY DIFFICULTY LEVEL (1 EASIEST, 5 HARDEST).

Level	n	Acc. (%)
1	43	51.2
2	90	31.1
3	105	24.8
4	128	8.6
5	134	5.2

TABLE VI
MGSM BY LANGUAGE.

Language	n	Acc. (%)
en	250	42.8
es	250	32.0
fr	250	28.4

TABLE VII

GGUF QUANTIZATION LADDER. PERPLEXITY IS MEASURED ON A HELD-OUT SLICE OF THE TRAINING DISTRIBUTION AT 512-TOKEN CONTEXT; LOWER IS BETTER. THE SUB-8-BIT FILES USE AN IMPORTANCE MATRIX CALIBRATED ON THAT DISTRIBUTION. Q5_K_M WAS RELEASED WITHOUT A PERPLEXITY MEASUREMENT.

Quantization	Size	RAM	Perplexity	vs. Q8_0
IQ4_XS	1.11 GB	1.6 GB	2.5347	+2.0%
Q4_0	1.12 GB	1.6 GB	2.5244	+1.6%
Q4_K_M	1.19 GB	1.7 GB	2.4907	+0.25%
Q5_K_M	1.31 GB	1.8 GB	—	—
Q8_0 (reference)	1.87 GB	2.4 GB	2.4844	—

inference entirely on a single 4 GB laptop GPU, at a peak of 1.81 GB of VRAM and in about 100 minutes. The resulting model is capable on science and general knowledge and weak on competition mathematics, advanced computer science, and strict instruction following, and its behavior is described well only by full-set measurements: a fifty-item GSM8K probe overstated the real figure by thirteen points, and self-consistency proved a cheaper reasoning lever than the model’s own tool-use pathway.

We do not claim that AVA-v2 surpasses its base or any larger model. The contribution is a complete and reproducible account of what low-resource adaptation costs and yields, and a reminder that the reliability of a small model’s reported numbers depends on how many questions produced them. For readers working under the same constraints, the released artifacts and the full-set measurements are meant to make the trade-offs concrete.

APPENDIX

a) Environment.: Training and evaluation ran on a single NVIDIA RTX A2000 Laptop GPU with 4 GB of memory, under Windows 11. Adaptation uses PEFT 0.18.1 with a 4-bit NF4 base and scaled dot-product attention; the base checkpoint reports transformers 4.57. Evaluation serves the Q8_0 GGUF with llama.cpp build 69c28f1. The random seed is 42. The complete pinned environment, corpus builders, training configuration, and evaluation harness are in the repository.

b) Decoding.: Multiple-choice scoring reads label-token probabilities and takes the arg-max. Generative decoding is greedy at temperature 0 with the Qwen ChatML-style template in non-thinking mode. Self-consistency uses five samples at temperature 0.7 with nucleus sampling and a majority vote over the extracted numeric answer.

c) Artifacts.: Adapter:

<https://huggingface.co/NAME0x0/AVA-v2>. Merged weights:

<https://huggingface.co/NAME0x0/AVA-v2-merged>. GGUF ladder: <https://huggingface.co/NAME0x0/AVA-v2-GGUF>. Base model: <https://huggingface.co/Qwen/Qwen3.5-2B>. Code and full evaluation report: <https://github.com/NAME0x0/AVA>.

d) Citing this work.:

```
@misc{mumtaz2026avav2,
  title = {AVA-v2: Reasoning on a 4 GB Laptop GPU},
  author = {Muhammad Afsah Mumtaz},
  year = {2026},
  url = {https://github.com/NAME0x0/AVA}
}
```

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